

HAT-P-36b

R-1216

Benchmark run · workflow W8-benchmark-deep · record deep-bench_HATP-36_220524 · created 2026-07-31T08:19:11+00:00

BENCHMARK: MARGINAL

Results

Mid-Transit Time (Tmid)	2459723.7514 +/- 0.0038 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.156 +/- 0.011
Transit depth (Rp/Rs)^2	2.43 +/- 0.33 [%]
Orbital Inclination (inc)	84.42 +/- 2.62
Airmass coefficient 1 (a1)	1.023 +/- 0.015
Airmass coefficient 2 (a2)	-0.018 +/- 0.013
Scatter in the residuals of the lightcurve fit is	1.52 %
Best Comparison Star	#1 - [166, 309]
Optimal Aperture	3.62
Optimal Annulus	10.57
Transit Duration (day)	0.0917 +/- 0.0081

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.156 $\pm$ 0.011 vs 0.126 (deviation 2.27 σ)
Duration fit vs published	0.0917 d vs 0.0925 d (deviation 0.08 σ)
Mid-time O-C	-11.6 min
Depth SNR	11.8
Residual scatter	1.52 %

Light curve

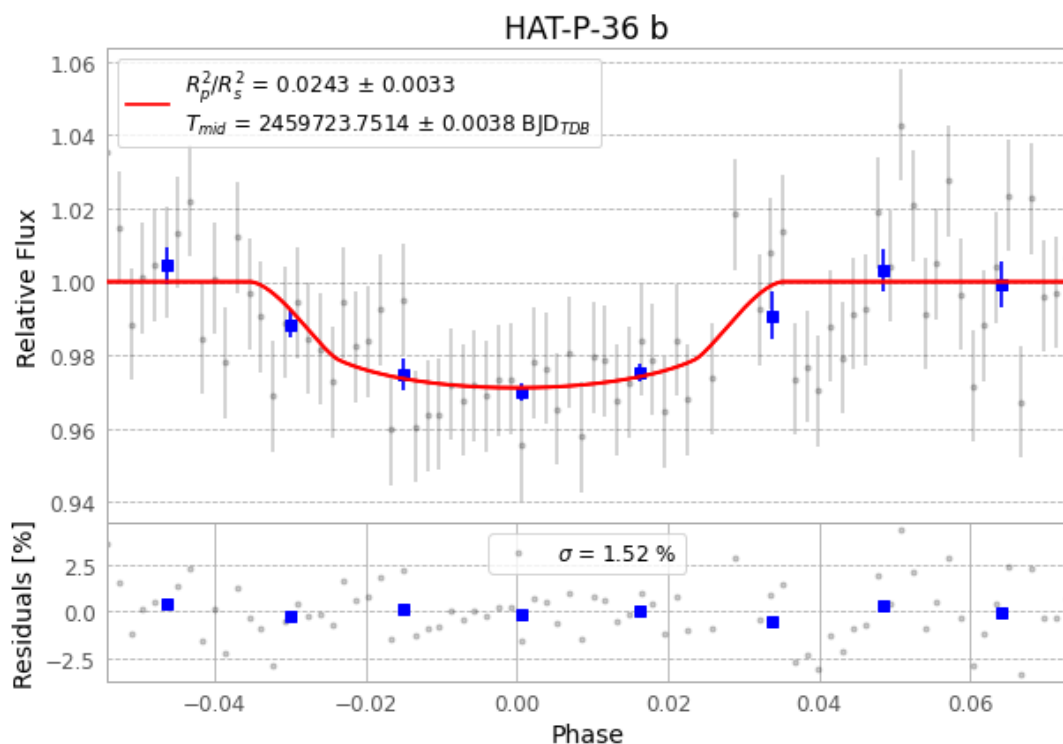


Figure 1 — FinalLightCurve_HAT-P-36 b_2022-05-23.png (SHA-256 e2a3a108a31cec08...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [310, 165], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 12e9d4d18741c5f0...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

81 FITS frames (53 MB), HATP-36220524041513.FITS → HATP-36220524081812.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-HATP-36-220524.log (0be1ad0a53da...), FinalLightCurve_HAT-P-36 b_2022-05-23.png (e2a3a108a31c...), FinalLightCurve_HAT-P-36 b_2022-05-23.csv (5baf01a5f5e5...)

Compute resources

Wall time 145.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-31 08:19Z.